Plots with Feature Values

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Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
library(stringr)
library(ggplot2)
library(ggrepel)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggExtra)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(ggpubr)
```

Load Data

Load csv file with quantitative feature values for sign sequences per object. Comparative sample: A comparative sample of feature values for languages across the world is derived from the Text Data Diversity (TeDDi) sample (Moran et al 2022).

SIGNBASE

```
# load signbase estimations for strings on objects
sign.data <- read.csv("~/Github/UpperPalCombinatorics/results/signBase_features.csv")
# select Aurignacian objects
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]
```

```r
# select columns for direct comparison with TeDDi sample
sign.data.selected <- select(sign.data, c('archaeological_unit', 'coding_clean',
                                          'num.char', 'huni.chars', 'hrate.chars',
                                 'ttr.chars', 'rm.chars'))
# change column names
colnames(sign.data.selected) <- c('subcorpus', 'sequence', 'num.char','huni.chars', 'hrate.chars',
                         'ttr.chars', 'rm.chars')
```

CDLI

```r
# read feature values for proto-cuneiform from CDLI
# Uruk V
cuneiV.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukV_features.csv")

# add column with subcorpus information
cuneiV.data$subcorpus <- rep("Cuneiform (Uruk V)", nrow(cuneiV.data))

# select columns for direct comparison with other samples
cuneiV.data.selected <- select(cuneiV.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiV.data.selected)[names(cuneiV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk IV
cuneiIV.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukIV_features.csv")

# add column with subcorpus information
cuneiIV.data$subcorpus <- rep("Cuneiform (Uruk IV)", nrow(cuneiIV.data))

# select columns for direct comparison with other sample
cuneiIV.data.selected <- select(cuneiIV.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiIV.data.selected)[names(cuneiIV.data.selected) == "signs.expand.vec"] <- "sequence"

# Uruk III
cuneiIII.data <- read.csv("~/Github/UpperPalCombinatorics/results/cuneiform_UrukIII_features.csv")

# add column with subcorpus information
cuneiIII.data$subcorpus <- rep("Cuneiform (Uruk III)", nrow(cuneiIII.data))

# select columns for direct comparison with other sample
cuneiIII.data.selected <- select(cuneiIII.data, c('subcorpus', 'signs.expand.vec',
                                              'num.char', 'huni.chars', 'hrate.chars',
                                              'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(cuneiIII.data.selected)[names(cuneiIII.data.selected) == "signs.expand.vec"] <- "sequence"
```

TeDDI

```r
# read feature values for character sequences of the TeDDi sample
teddi.data <- read.csv("~/Github/UpperPalCombinatorics/results/teddi_features.csv")
# add column with subcorpus information
teddi.data$subcorpus <- rep("TeDDi", nrow(teddi.data))
# select columns
teddi.selected <- select(teddi.data, c('subcorpus','text', 'num.char','huni.chars', 'hrate.chars',
                            'ttr.chars', 'rm.chars'))
# rename column with sign sequences
names(teddi.selected)[names(teddi.selected) == "text"] <- "sequence"
```

Preprocessing

```r
# combine the data frames
estimations.df <- rbind(sign.data.selected, cuneiV.data.selected,
                        cuneiIV.data.selected, cuneiIII.data.selected,
                        teddi.selected)
nrow(estimations.df)
```

```
## [1] 2991
```

```r
# select sign strings of certain length
#estimations.df <- estimations.df[estimations.df$num.char > 10,]

# select only certain subcorpora for plotting
estimations.df.selected <- estimations.df[estimations.df$subcorpus == "Aurignacien"|
                                          #estimations.df$subcorpus == "Aurignacien/Gravettian"|
                                          estimations.df$subcorpus == "Cuneiform (Uruk V)"|
                                          estimations.df$subcorpus == "Cuneiform (Uruk IV)"|
                                          estimations.df$subcorpus == "Cuneiform (Uruk III)"|
                                          estimations.df$subcorpus == "TeDDi", ]
```

Plots

Get subset of data points to plot labels.

```r
labels.df <- estimations.df.selected[estimations.df.selected$num.char == 11 &
                                     estimations.df.selected$subcorpus != "TeDDi" &
                                     estimations.df.selected$subcorpus != "Cuneiform (Uruk III)",]
```

Entropy rate vs. unigram entropy

```r
huni.hrate.chars.plot <- ggplot(estimations.df.selected, aes(x = huni.chars, y =  hrate.chars,
                                colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  geom_label_repel(data = labels.df,
```

```
                 aes(label = sequence),
    position = position_nudge_repel(x = 0, y = 0.5),
    size = 3,
    show.legend = FALSE) +
    theme(legend.position = c(.8, .25))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                   type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

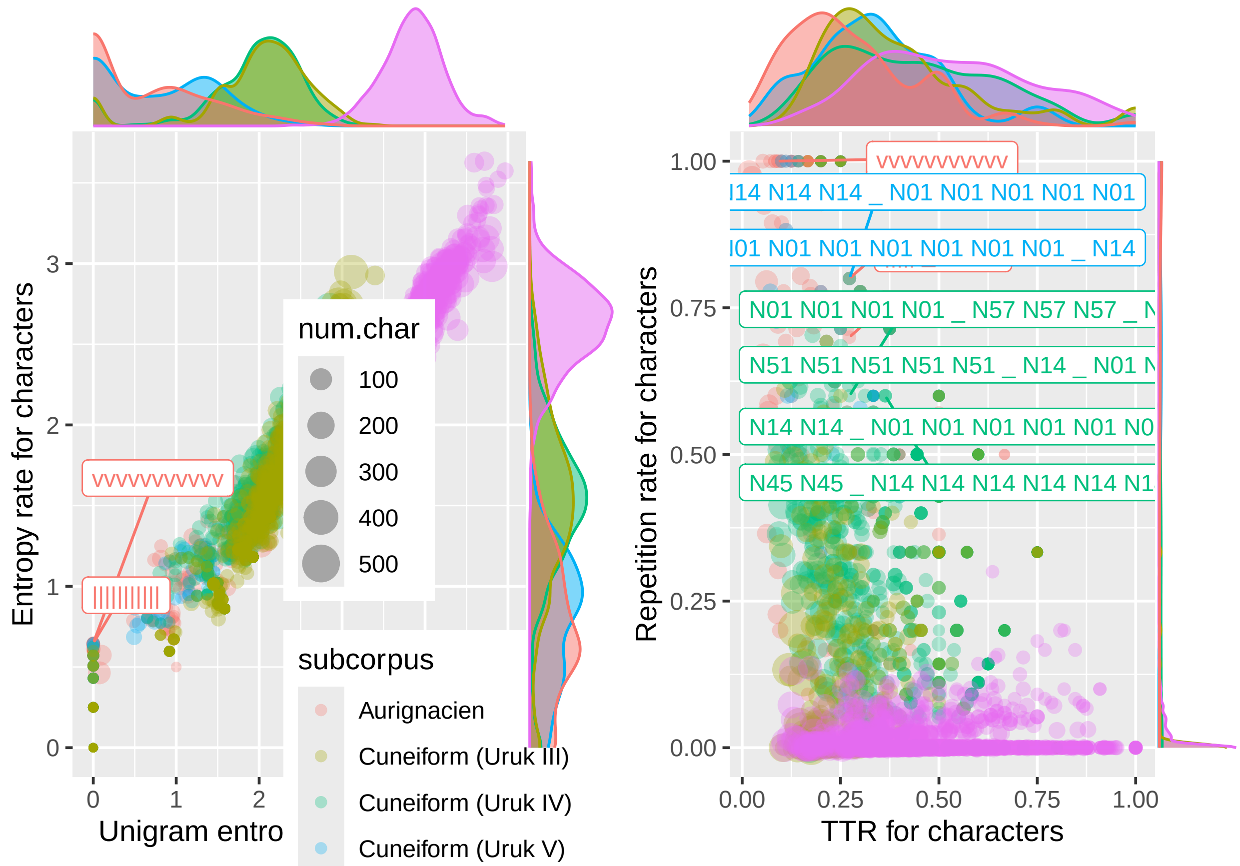
```
ttr.rm.chars.plot <- ggplot(estimations.df.selected, aes(x = ttr.chars, y = rm.chars,
                                   colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
  theme(legend.position = "none") +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  geom_label_repel(data = labels.df,
                 aes(label = sequence),
    position = position_nudge_repel(x = 0.25, y = 0),
    size = 3,
    show.legend = FALSE)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                   type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```

```
## Warning: ggrepel: 53 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps

## Warning: ggrepel: 45 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_chars.pdf",
       combined.plot, dpi = 300, width = 12, height = 6, device = cairo_pdf)
```

```
## Warning: ggrepel: 53 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps

## Warning: ggrepel: 45 unlabeled data points (too many overlaps). Consider
## increasing max.overlaps
```

Swabian Jura Specific Plots

Comparisons of different caves of the Swabian Jura.

Entropy rate vs. unigram entropy

```
huni.hrate.chars.plot <- ggplot(sign.data, aes(x = huni.chars, y = hrate.chars,
                                        colour = site_name, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
```

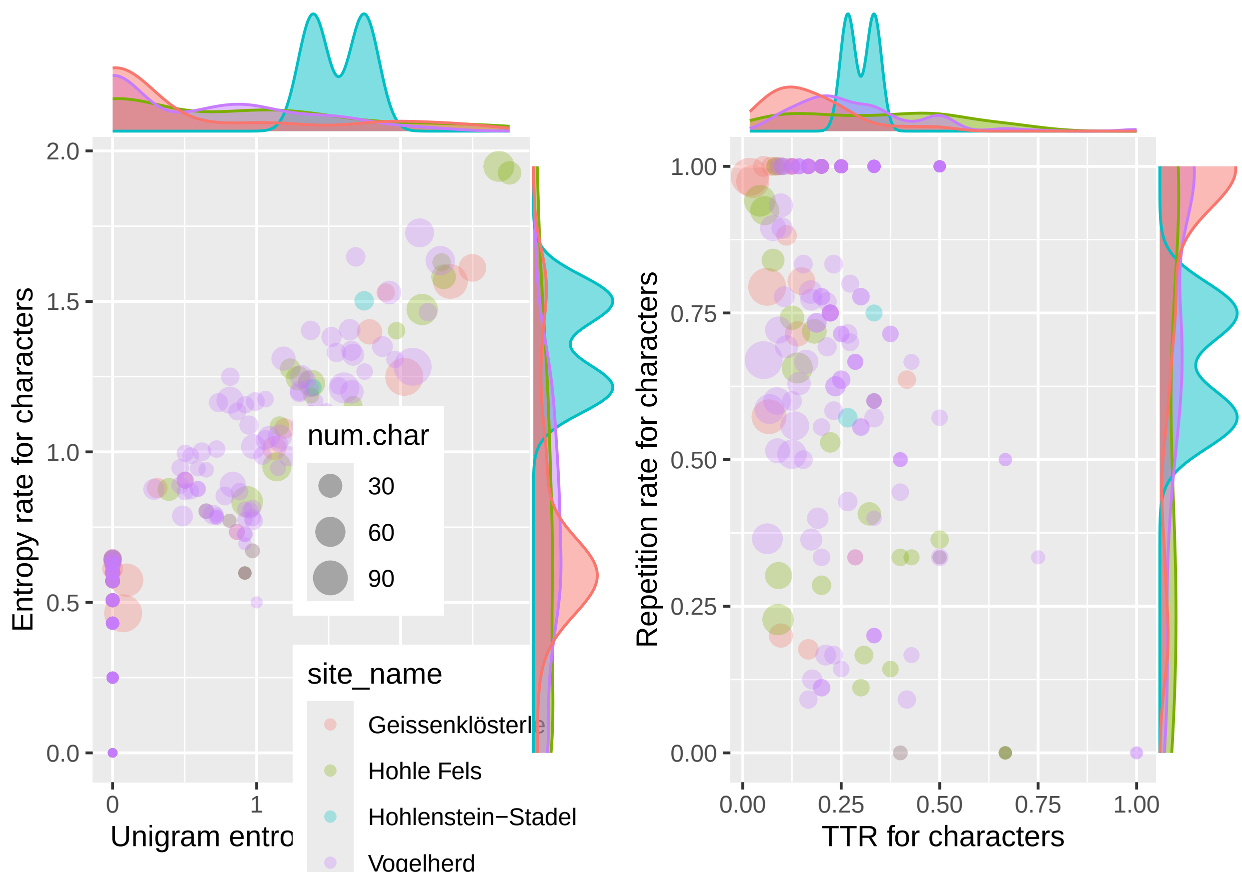
```
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                     type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(sign.data, aes(x = ttr.chars, y = rm.chars,
                                           colour = site_name, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                 type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_Aurignacian.pdf",
       combined.plot, dpi = 300, width = 20, height = 10, device = cairo_pdf)
```

TeDDi Specific Plots

Comparisons of different writing systems in TeDDi.

Entropy rate vs. unigram entropy

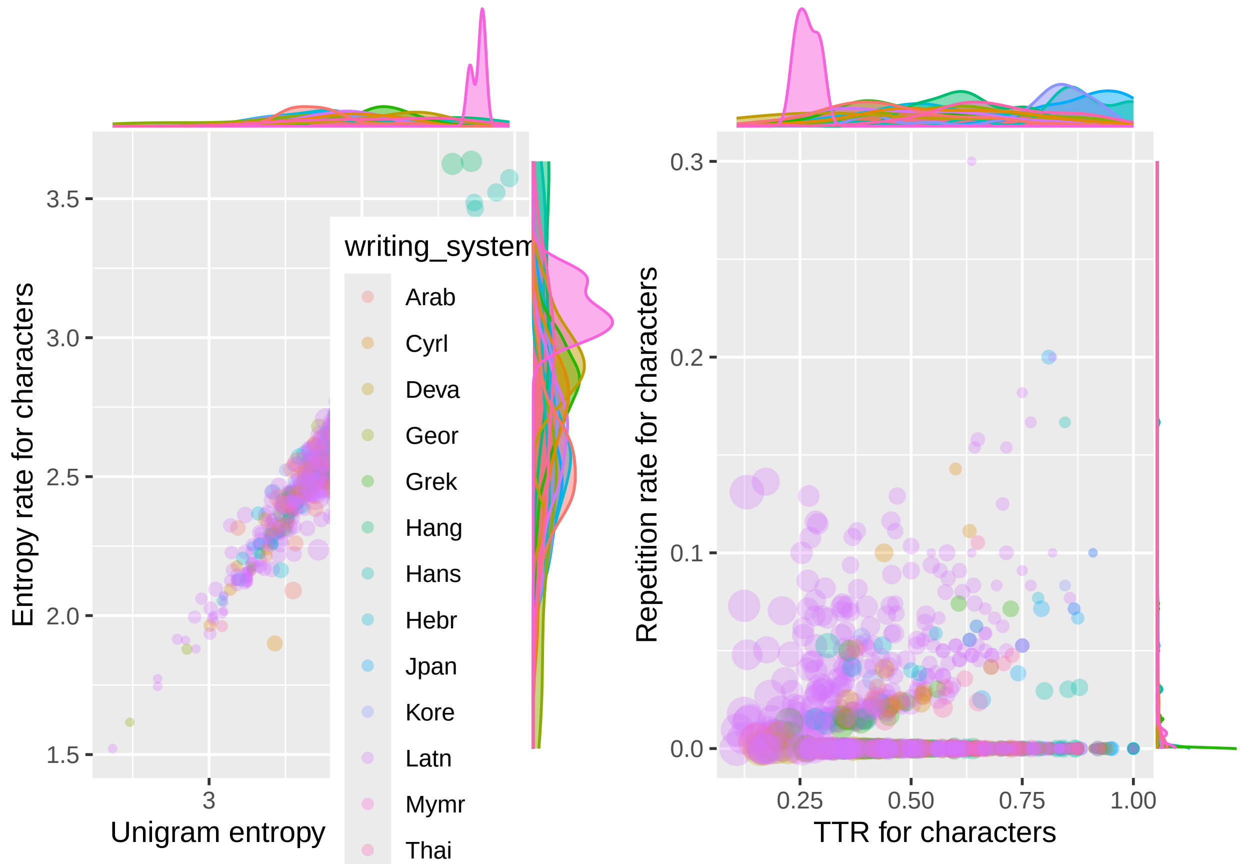
```
huni.hrate.chars.plot <- ggplot(teddi.data, aes(x = huni.chars, y = hrate.chars,
                                    colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(teddi.data, aes(x = ttr.chars, y = rm.chars,
                                    colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/writingSystem_plot.pdf",
       combined.plot, dpi = 300, width = 25, height = 15, device = cairo_pdf)
```